

INSURANCE CLAIMS ANALYTICS

Claims Operations | End-to-End Data Analytics Project

Excel

SQL / Snowflake

Python / Databricks

Power BI

Streamlit

Machine Learning

Presented by

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Executive Summary

Scope

This project analyzes insurance claims data to understand claim volume, financial exposure, customer characteristics, risk segments, incident severity and operational reporting patterns.

Approach

An end-to-end analytics workflow: data profiling and quality assessment, SQL-based cleaning and transformation, Python analysis, business intelligence reporting and predictive modeling.

Delivery

The final solution was delivered through Power BI and Streamlit to provide interactive, business-friendly analytical insights.

10,000

Insurance Claims

₹165.64M

Total Claim Amount

6

Insurance Types

1,200

Agents

3

Risk Segments

Business Problem

Insurance companies generate large volumes of claims data containing customer, policy, incident, financial and operational information.

Without systematic analysis, it is difficult to:

- Understand overall claim volume and financial exposure.
- Identify patterns across insurance types and risk segments.
- Understand incident severity and claim-value distribution.
- Monitor reporting delays and operational patterns.
- Compare claim activity across agents and vendors.
- Use historical claim characteristics to estimate future claim amounts.

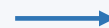
PROBLEM STATEMENT

How can insurance claims data be transformed into actionable insights for understanding claim exposure, risk, severity, operations and claim amount prediction?

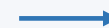
Objective

Transform raw insurance claims data into a structured analytical solution that supports reporting, pattern identification and predictive analysis.

Raw Claims Data



Structured Analysis



Business Decisions

Business Objectives

Six analytical objectives drive the scope of this project

01

Claims & Financial Analysis

Measure claim volume, total claim exposure, average and median claim amount and premium patterns.

02

Risk & Severity Analysis

Understand the distribution of risk segments and incident severity across insurance claims.

03

Customer & Demographic Analysis

Analyze customer age, marital status, employment status and other demographic attributes.

04

Operational Analysis

Analyze reporting delays and examine claim activity across agents and vendors.

05

Diagnostic Analysis

Identify relationships between risk, severity, insurance type, injury, police reports and claim amounts.

06

Predictive Analysis

Build regression models to estimate claim amounts using historical claim and customer characteristics.

Dataset Overview

Three related datasets form the analytical foundation

Dataset	Rows	Columns	Purpose
Insurance	10,000	38	Main claims-level dataset
Employee	1,200	10	Agent reference information
Vendor	600	7	Vendor reference information

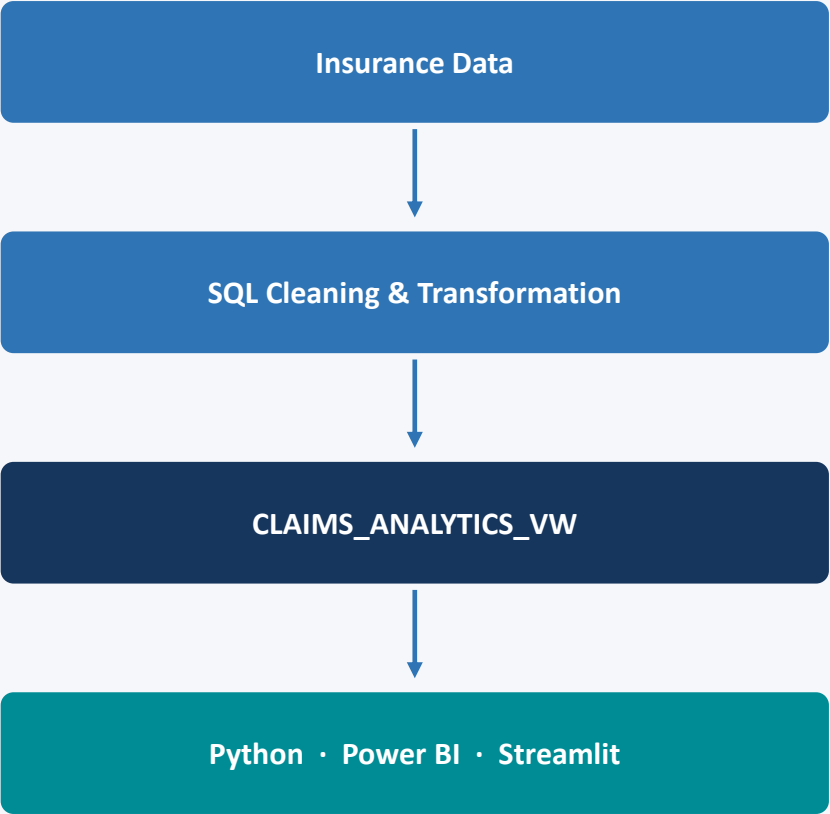
Total Records Across Datasets

11,800

Transaction Period

June 2020 – June 2021 · 13 months of claims activity

Main Analytical Path



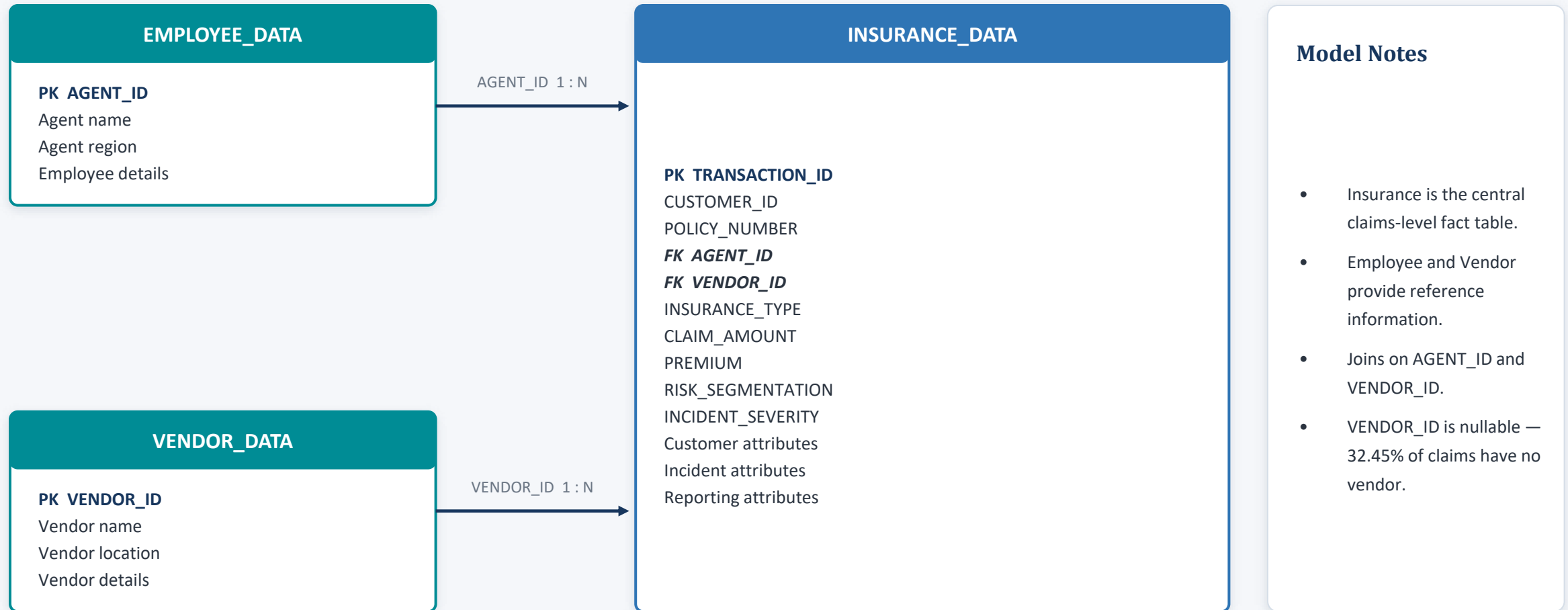
Data Dictionary — Insurance Claims

Key analytical fields used throughout the project

Column	Description	Column	Description
TRANSACTION_ID	Unique identifier for each claim transaction	RISK_SEGMENTATION	Risk segment of the claim / customer
CUSTOMER_ID	Unique customer identifier	INCIDENT_SEVERITY	Severity classification of the incident
POLICY_NUMBER	Insurance policy identifier	ANY_INJURY	Whether injury was reported
TXN_DATE_TIME	Transaction date and time	POLICE_REPORT_AVAILABLE	Availability of police report
POLICY_EFF_DT	Policy effective date	AGENT_ID	Agent associated with the claim
LOSS_DT	Date of loss / incident	VENDOR_ID	Vendor associated with the claim
REPORT_DT	Date the incident was reported	MARITAL_STATUS	Customer marital status
INSURANCE_TYPE	Type of insurance	EMPLOYMENT_STATUS	Customer employment status
CLAIM_AMOUNT	Monetary claim amount	HOUSE_TYPE	Customer house type
PREMIUM	Insurance premium amount	SOCIAL_CLASS	Customer social classification
AGE	Customer age	CUSTOMER_EDUCATION_LEVEL	Customer education level
TENURE	Customer / policy tenure		

The complete field-level dictionary is provided in Appendix A.

Data Model — Entity Relationship Diagram



Technology Stack

DATA PROFILING

Microsoft Excel

- Data profiling
- Missing-value analysis
- Duplicate checks
- Data quality assessment

DATA WAREHOUSE

SQL / Snowflake

- Data storage
- SQL cleaning
- Data transformation
- Analytical views

DATA ANALYSIS

Python / Databricks

- Pandas · NumPy
- Matplotlib · Seaborn
- Exploratory analysis
- Diagnostic analysis

BUSINESS INTELLIGENCE

Power BI

- KPI reporting
- Interactive dashboards
- Slicers
- Business visualizations

APPLICATION

Streamlit

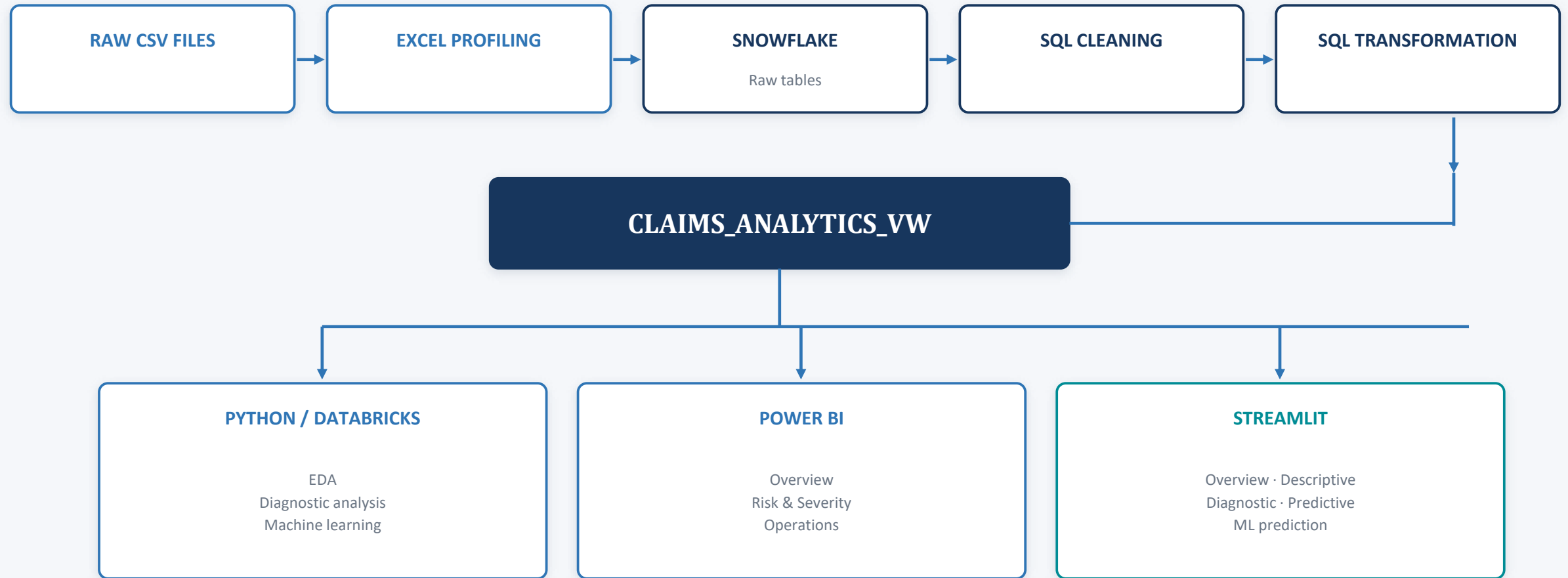
- Interactive Python app
- Filters & dynamic KPIs
- Analytics pages
- ML prediction interface

MACHINE LEARNING

Scikit-learn

- Linear Regression
- Random Forest
- MAE · RMSE · R^2
- Model evaluation

End-to-End Project Architecture



The analytical view acts as the common analytical layer for Python, Power BI and Streamlit — eliminating duplication of cleaning logic across tools.

Data Quality Assessment

0

Duplicate Rows

0

Invalid Agent References

0

Invalid Vendor References

0

Negative Claim / Premium

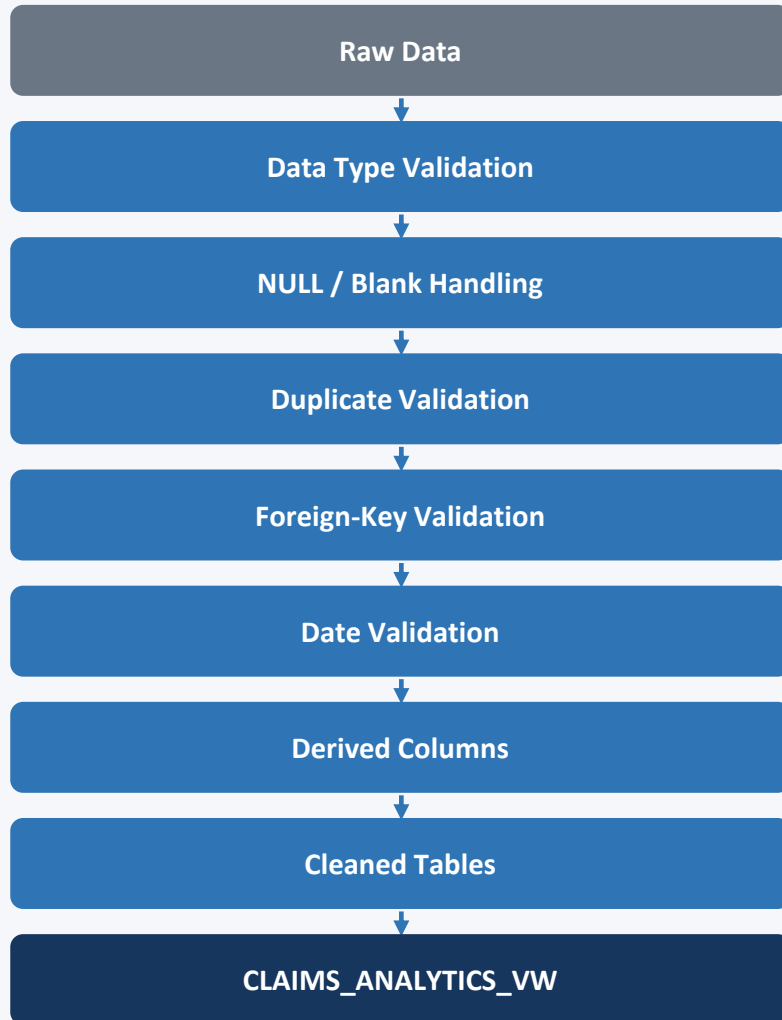
Major Missing-Value Findings

ADDRESS_LINE2		8,505 (85.05%)
VENDOR_ID		3,245 (32.45%)
AUTHORITY_CONTACTED		1,945 (19.45%)
CUSTOMER_EDUCATION_LEVEL		529 (5.29%)
CITY		54 (0.54%)
INCIDENT_CITY		46 (0.46%)

Validation Checks Passed

- ✓ No full-row duplicates identified.
- ✓ No invalid Agent_ID references identified.
- ✓ No invalid non-null Vendor_ID references identified.
- ✓ Reporting delay remained between 0 and 5 days.
- ✓ No policy effective date occurred after the loss date.
- ✓ No report date occurred before the loss date.

SQL Data Cleaning & Transformation



Derived Fields Created

- REPORTING_DELAY_DAYS
- CLAIM_TO_PREMIUM_RATIO
- AGE_GROUP
- TENURE_GROUP
- CLAIM_AMOUNT_BUCKET
- REPORTING_CATEGORY
- CLAIM_VALUE_CATEGORY
- LOSS_YEAR
- LOSS_MONTH
- LOSS_MONTH_NAME

Data Security

Sensitive fields such as SSN, account number and routing number were excluded from the downstream analytical dataset.

Why a View?

All cleaning and derivation logic lives once in SQL, so Python, Power BI and Streamlit consume identical, consistent data.

Descriptive Analysis

10,000

Total Claims

₹165.64M

Total Claim Amount

₹16,563.83

Average Claim

₹7,000

Median Claim

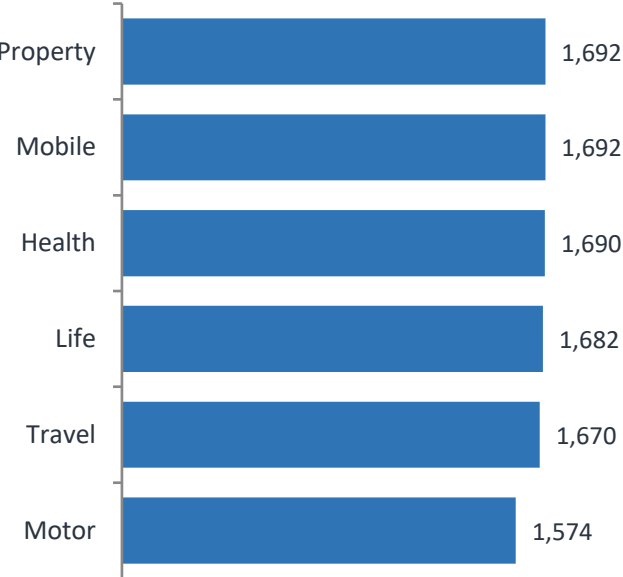
₹88.51

Average Premium

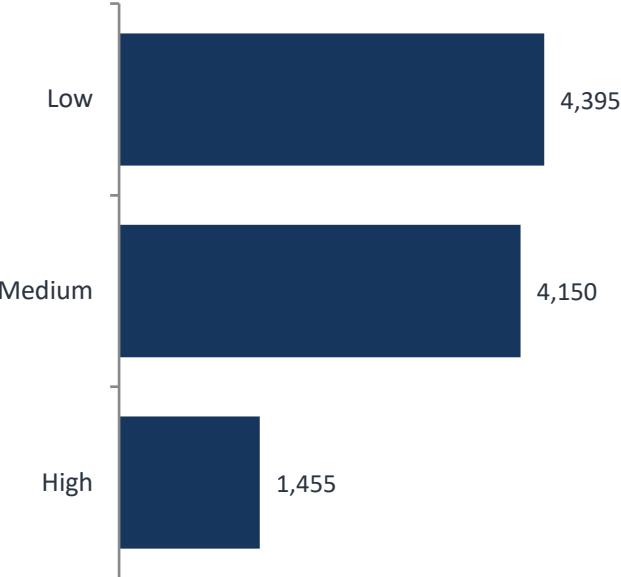
3.21 days

Avg Reporting Delay

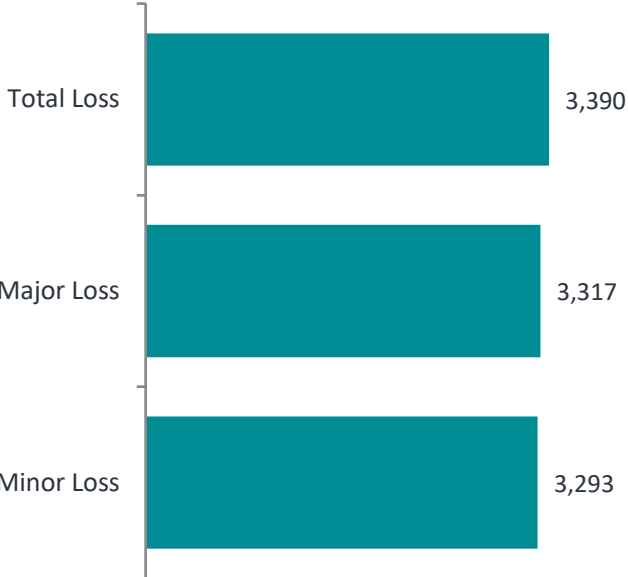
Claims by Insurance Type



Claims by Risk Segment



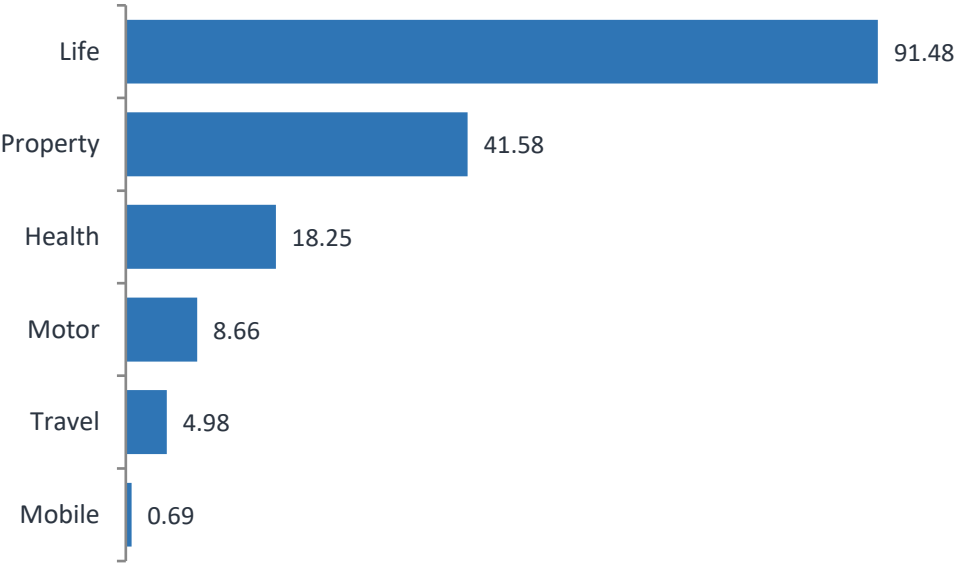
Claims by Incident Severity



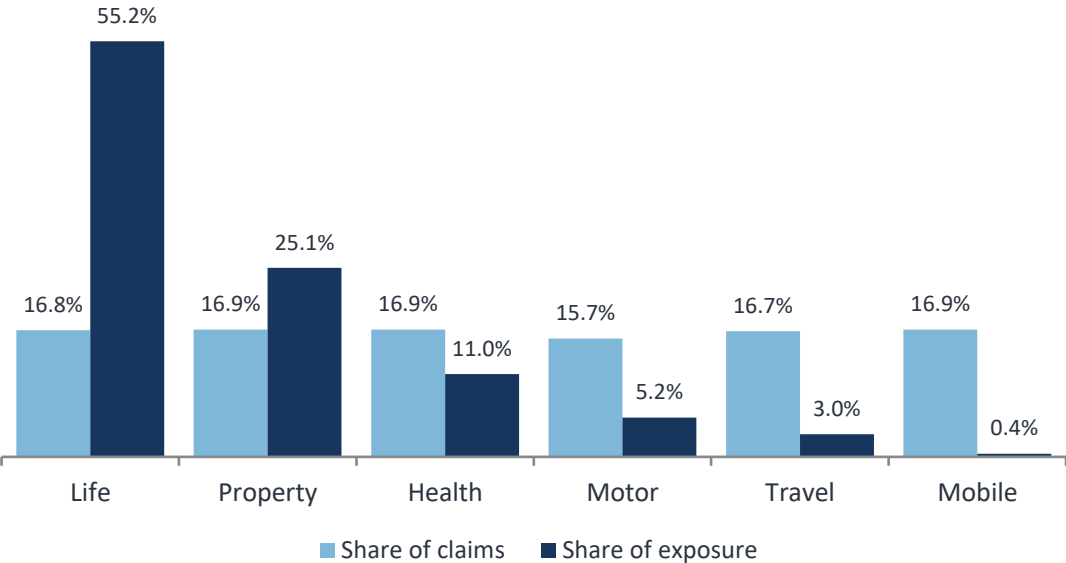
Claim Exposure by Insurance Type

Claim volume is split almost evenly across six products — claim value is not

Total Claim Amount by Insurance Type (₹ Million)



Share of Claims vs Share of Exposure



55.2%

of total exposure comes from Life claims, which are only 16.8% of the claim count

80.3%

of total exposure sits in just two products: Life and Property

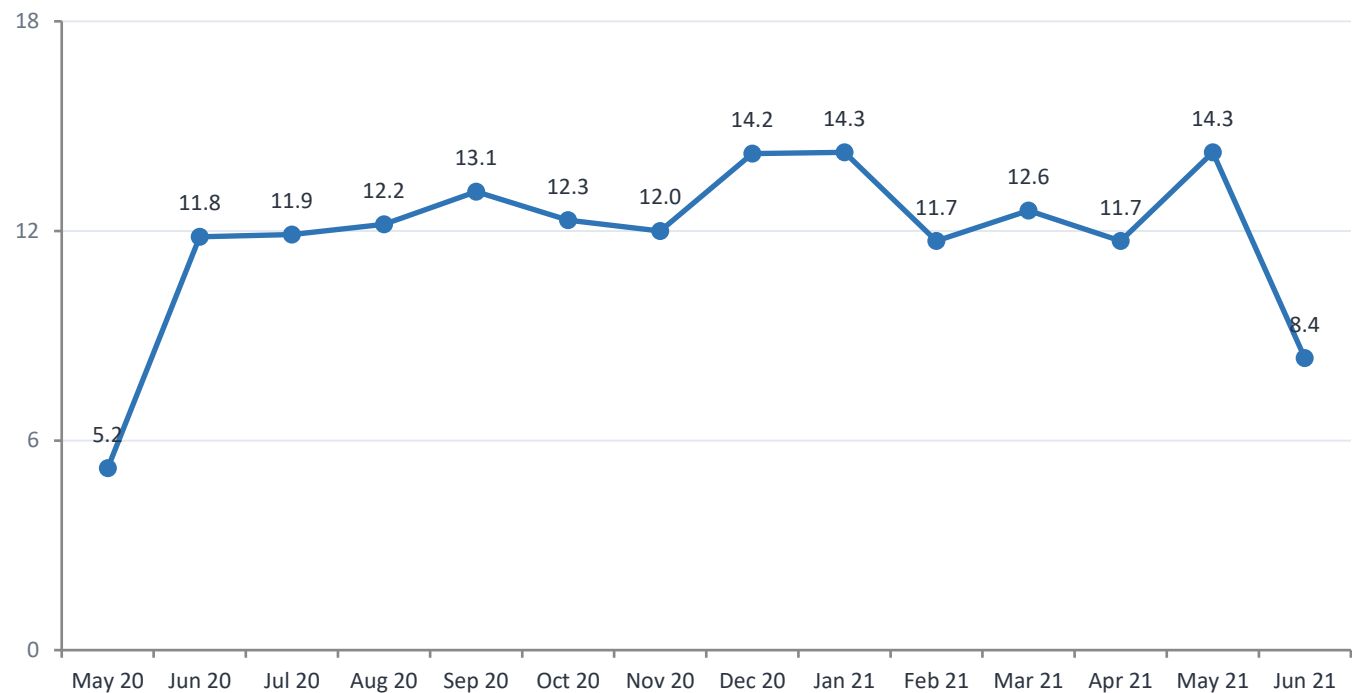
₹100,000

is the highest claim value — the top 10 claims are all Life policies

Monthly Trend & Top States

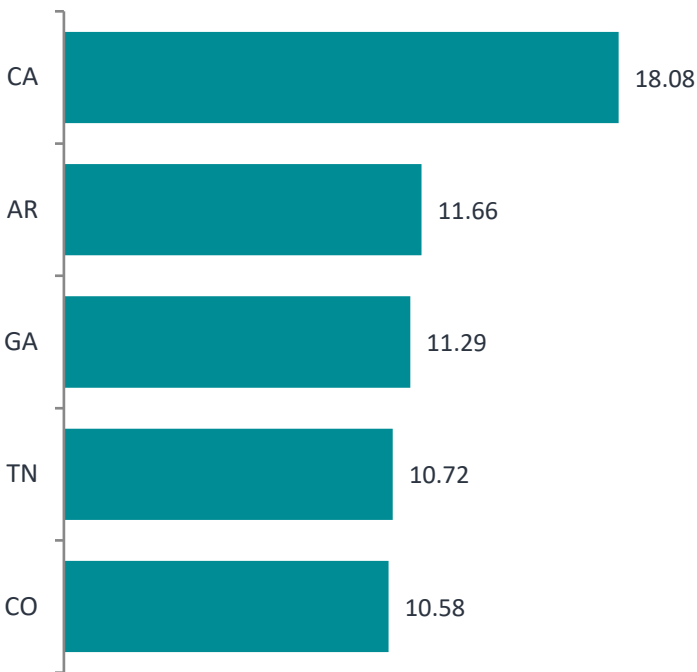
Claim amount by loss month, and where exposure is concentrated geographically

Monthly Claim Amount (₹ Million, by loss month)



Full months run at ₹11.7M–₹14.3M. May 2020 and Jun 2021 are lower, most likely because they are partial months.

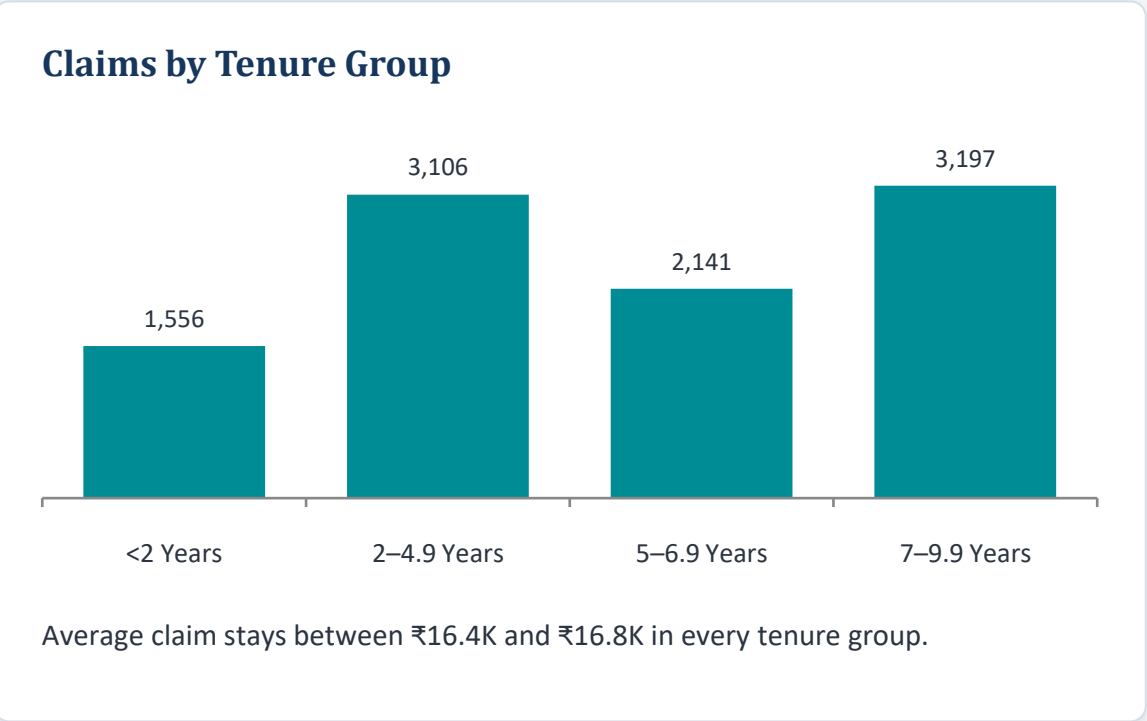
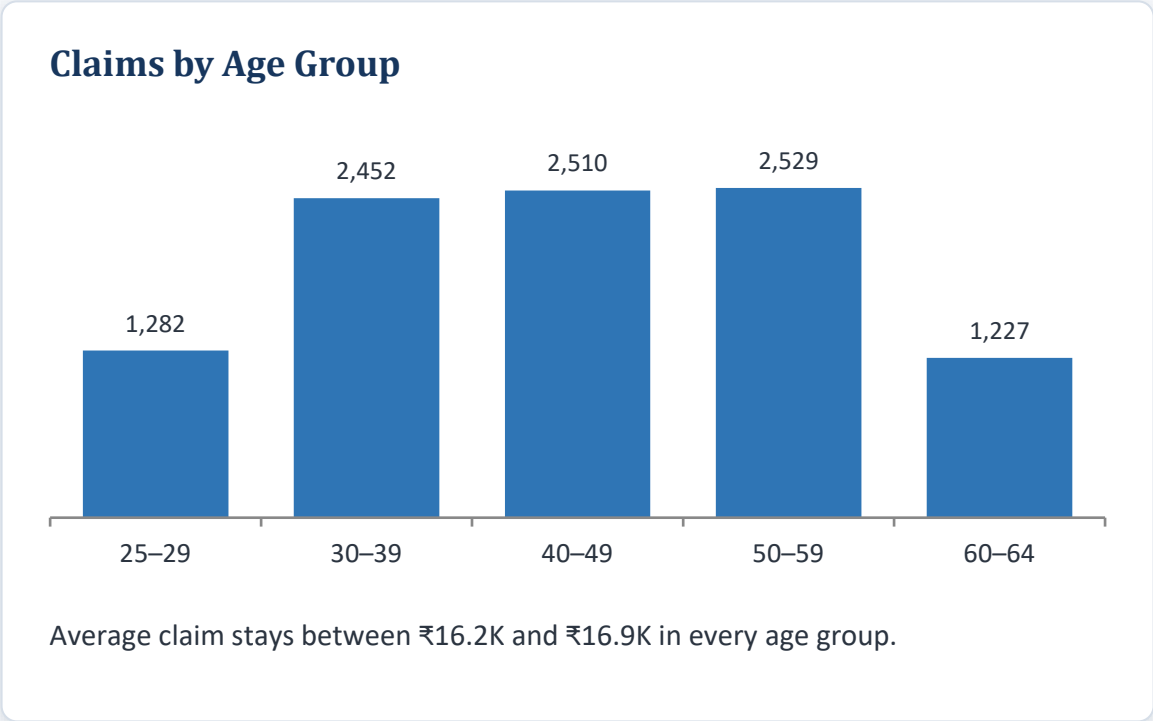
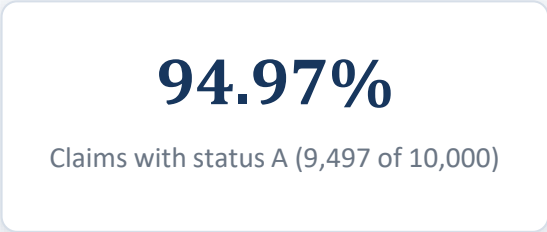
Top 5 States by Claim Amount (₹M)



California leads with 1,035 claims (10.4%) and ₹18.08M (10.9% of exposure), across 16 states in total.

Customer Profile & Claim Characteristics

Who files claims, and how those claims are recorded



Diagnostic Analysis

Examining relationships between claim attributes

QUESTION	VERDICT	EVIDENCE	WHAT IT MEANS
1 Does risk vary across incident severity?	PATTERN OBSERVED	Low 4,395 · Medium 4,150 · High 1,455	Shows how risk categories are distributed across severity levels.
2 Does risk profile differ by insurance type?	PATTERN OBSERVED	6 insurance types · 10,000 claims	Helps compare risk composition across insurance products.
3 Does claim amount vary by severity?	PATTERN OBSERVED	Average ₹16,563.83 · Median ₹7,000	Helps understand financial exposure across severity levels.
4 Does reporting delay relate to claim amount?	PATTERN OBSERVED	Delay 0–5 days · Average 3.21 days	Helps compare claim exposure across reporting timelines.
5 Do agents and vendors have different claim exposure?	PATTERN OBSERVED	Agent & vendor claim volume and exposure analyzed	Shows how claims are distributed across operational entities.

KEY TAKEAWAY Diagnostic analysis identifies relationships and patterns in claim data. It does not establish causation.

Risk, Severity & Insurance Type

Does risk vary by incident severity, or by insurance type?

Risk Segment × Incident Severity

Risk	Minor Loss	Major Loss	Total Loss
Low	1,453 · 33.1%	1,439 · 32.7%	1,503 · 34.2%
Medium	1,347 · 32.5%	1,416 · 34.1%	1,387 · 33.4%
High	493 · 33.9%	462 · 31.8%	500 · 34.4%

Cells show number of claims · % of that risk segment. Each segment splits roughly one-third across the three severity levels.

Insurance Type × Risk Segment (% of claims)

Insurance Type	Low	Medium	High
Health	44.6%	41.1%	14.3%
Life	43.5%	41.6%	14.9%
Mobile	42.6%	42.6%	14.9%
Motor	45.5%	41.2%	13.3%
Property	42.5%	41.8%	15.7%
Travel	45.3%	40.7%	14.1%

KEY TAKEAWAY

Risk mix is nearly identical across all six insurance types, and severity splits evenly within every risk segment.

What Drives Claim Amount?

Average claim amount compared across every factor analysed in the notebook

Average Claim Amount: Lowest vs Highest Group

Factor	Lowest group	Highest group	Gap
Insurance type	Mobile · ₹407	Life · ₹54,386	133.7×
Age group	50–59 · ₹16,195	60–64 · ₹16,919	4.5%
Incident severity	Minor Loss · ₹16,232	Total Loss · ₹16,796	3.5%
Injury reported	No · ₹16,171	Yes · ₹16,733	3.5%
Reporting timing	Delayed · ₹16,488	Same day · ₹16,971	2.9%
Tenure group	<2 Years · ₹16,370	2–4.9 Years · ₹16,786	2.5%
Risk segment	Medium · ₹16,519	High · ₹16,825	1.9%
Police report	No · ₹16,389	Yes · ₹16,641	1.5%

KEY FINDING

134×

Life claims average ₹54,386 versus ₹407 for Mobile — the only large gap in the dataset.

Everything else is flat

Risk, severity, injury, police report, reporting delay, age and tenure shift the average claim by less than 5%. They describe claims but do not separate claim size.

Reporting, Agents & Vendors

Operational patterns in reporting behaviour and claim exposure

84.35%

Claims reported after the loss date

3.81 days

Average delay on delayed claims

₹16,971

Average same-day claim (delayed:
₹16,488)

32.45%

Claims with no vendor ID (avg claim
₹16,446)

Top 5 Agents by Claim Exposure

Agent	Claims	Total (₹)	Avg claim (₹)
AGENT00807	14	528,800	37,771
AGENT00679	13	489,000	37,615
AGENT00771	15	422,100	28,140
AGENT00125	10	400,400	40,040
AGENT00789	9	392,900	43,656

These agents handle 9–15 claims (average agent: 8.3) at above-average claim sizes. Exposure reflects workload and claim value, not performance.

Top 5 Vendors by Claim Exposure

Vendor	Claims	Total (₹)	Avg claim (₹)
VNDR00445	23	661,000	28,739
VNDR00453	18	635,000	35,278
VNDR00031	25	608,700	24,348
VNDR00591	21	598,000	28,476
VNDR00138	19	567,200	29,853

These vendors handle 18–25 claims (average vendor: 16.6) at above-average claim sizes.

Predictive Analysis — Claim Amount Prediction

Objective

Predict CLAIM_AMOUNT using historical customer, claim, risk, incident and reporting attributes.

TARGET VARIABLE

CLAIM_AMOUNT

Models & Evaluation

1 · Linear Regression

2 · Random Forest

MAE

Mean Absolute Error

RMSE

Root Mean Squared Error

R²

Coefficient of
Determination

Feature Set

Insurance Type

Age

Tenure

Risk Segmentation

Incident Severity

Marital Status

Employment Status

House Type

Social Class

Education Level

Any Injury

Police Report Available

Incident Hour

Reporting Delay

Predictive Modeling Approach

From selected features to evaluated regression models

14

Features (10 categorical + 4 numeric)

8,000

Training rows (80%)

2,000

Test rows (20%)

200

Trees in the Random Forest

1

Select Features

14 customer, claim, risk, incident and reporting attributes. Target: CLAIM_AMOUNT.

2

Split Data

80 / 20 train-test split with random_state = 42, giving 8,000 and 2,000 rows.

3

Preprocess

Numeric: median imputation. Categorical: most-frequent imputation and one-hot encoding.

4

Train Models

Linear Regression and Random Forest (200 trees), each inside one Pipeline.

5

Evaluate

MAE, RMSE and R^2 on the 2,000 test rows; the best model is selected.

KEY TAKEAWAY

Preprocessing and model live in a single Pipeline, so identical steps are applied to training and test data.

Predictive Model Performance

Model	MAE	RMSE	R ²
Linear Regression	6,935.15	11,976.84	0.7066
Random Forest	6,903.99	12,250.86	0.6930

Model Selection

Linear Regression was selected as the final model based on combined test performance.

Random Forest achieved a slightly lower MAE, but Linear Regression achieved a lower RMSE and a higher R². Linear Regression therefore provided the stronger overall performance for this dataset based on the selected evaluation metrics.

FINAL MODEL

Linear Regression

$R^2 = 0.7066$

≈ 71% of the variation in claim amount is explained by the selected features.

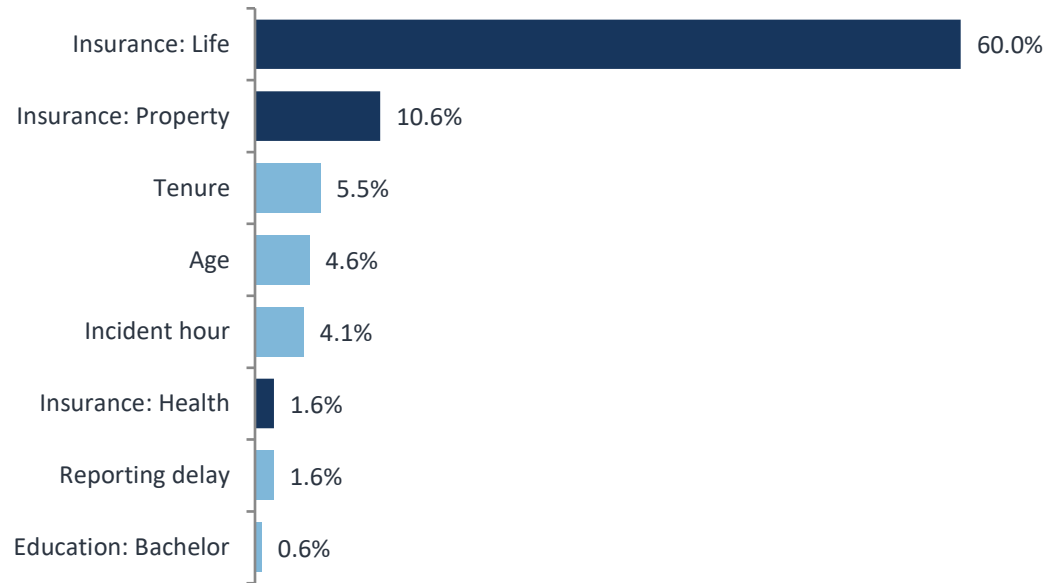
Interpretation Note

Predictions represent estimates of claim amount and should not be interpreted as guaranteed claim settlements.

What Drives the Predictions?

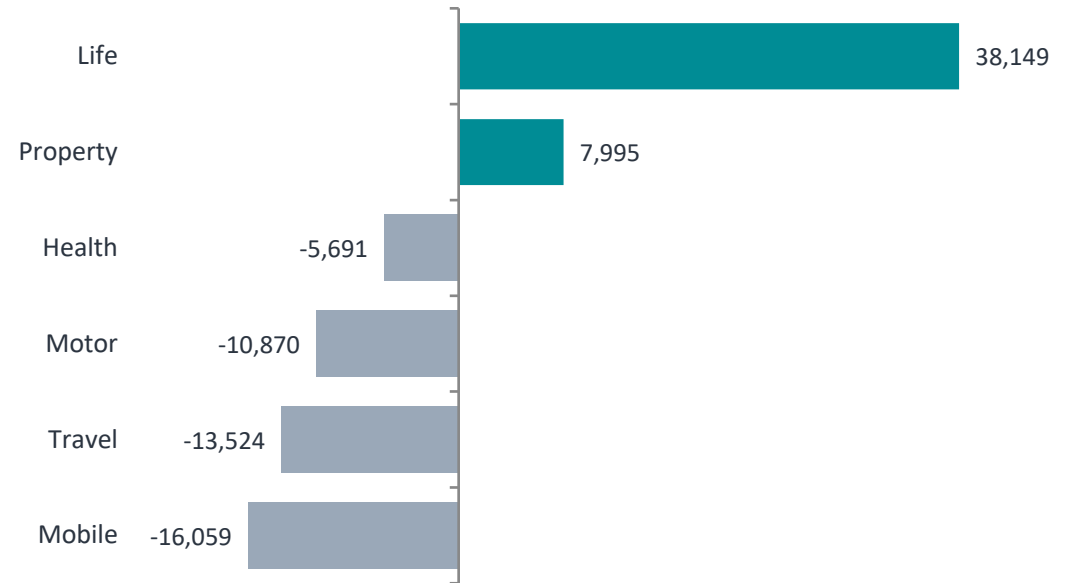
Random Forest feature importance and Linear Regression insurance-type coefficients

Random Forest — Top Feature Importance



Dark bars = insurance-type features. Every feature not shown is below 0.6%.

Linear Regression — Insurance Type Coefficients (₹)



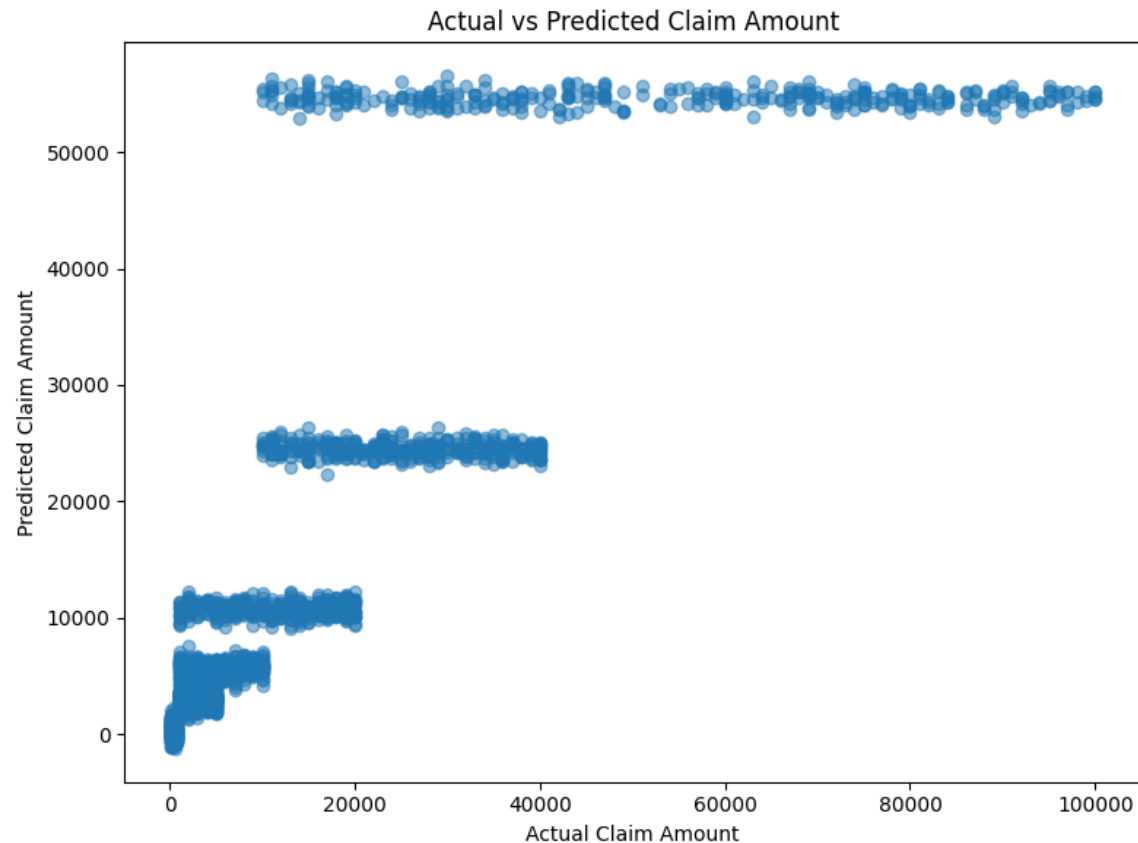
One-hot coefficients are relative to each other: Life lifts the prediction most, Mobile least.

KEY TAKEAWAY Insurance type is the dominant predictor in both models, consistent with the diagnostic findings.

Actual vs Predicted Claim Amount

Linear Regression on the 2,000-row test set

Test-Set Predictions (2,000 claims)



Predictions form bands

Predicted values sit in flat bands that line up with each insurance type's average claim: Life ₹54K, Property ₹25K, Health ₹11K.

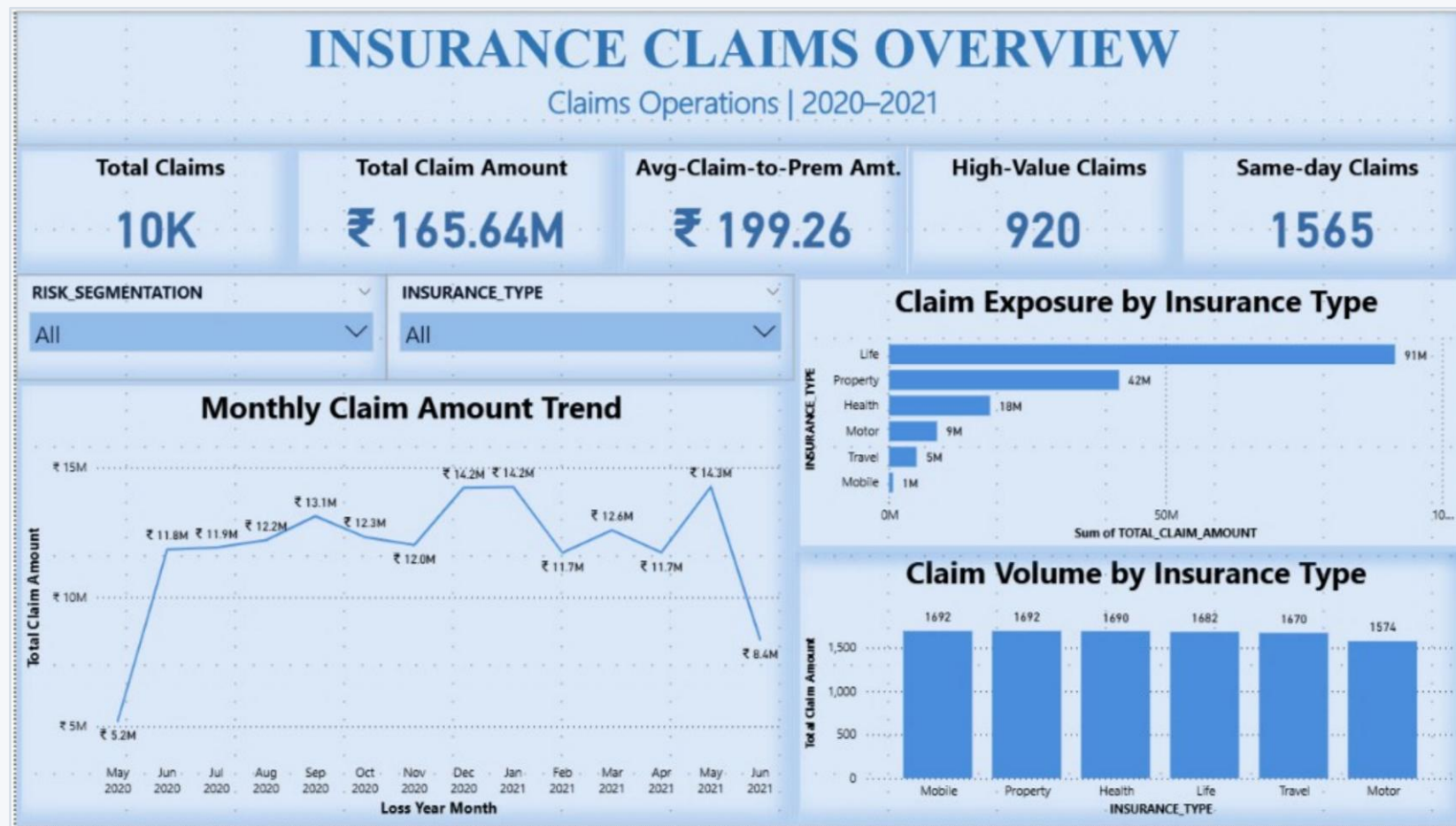
Typical error is large

MAE is ₹6,935 — about the size of the median claim (₹7,000). A ₹75,000 claim, for example, was predicted at ₹55,131.

Best use

Estimating exposure by product and segment for planning — not settling individual claims.

Power BI Dashboard 01 — Insurance Claims Overview



Purpose

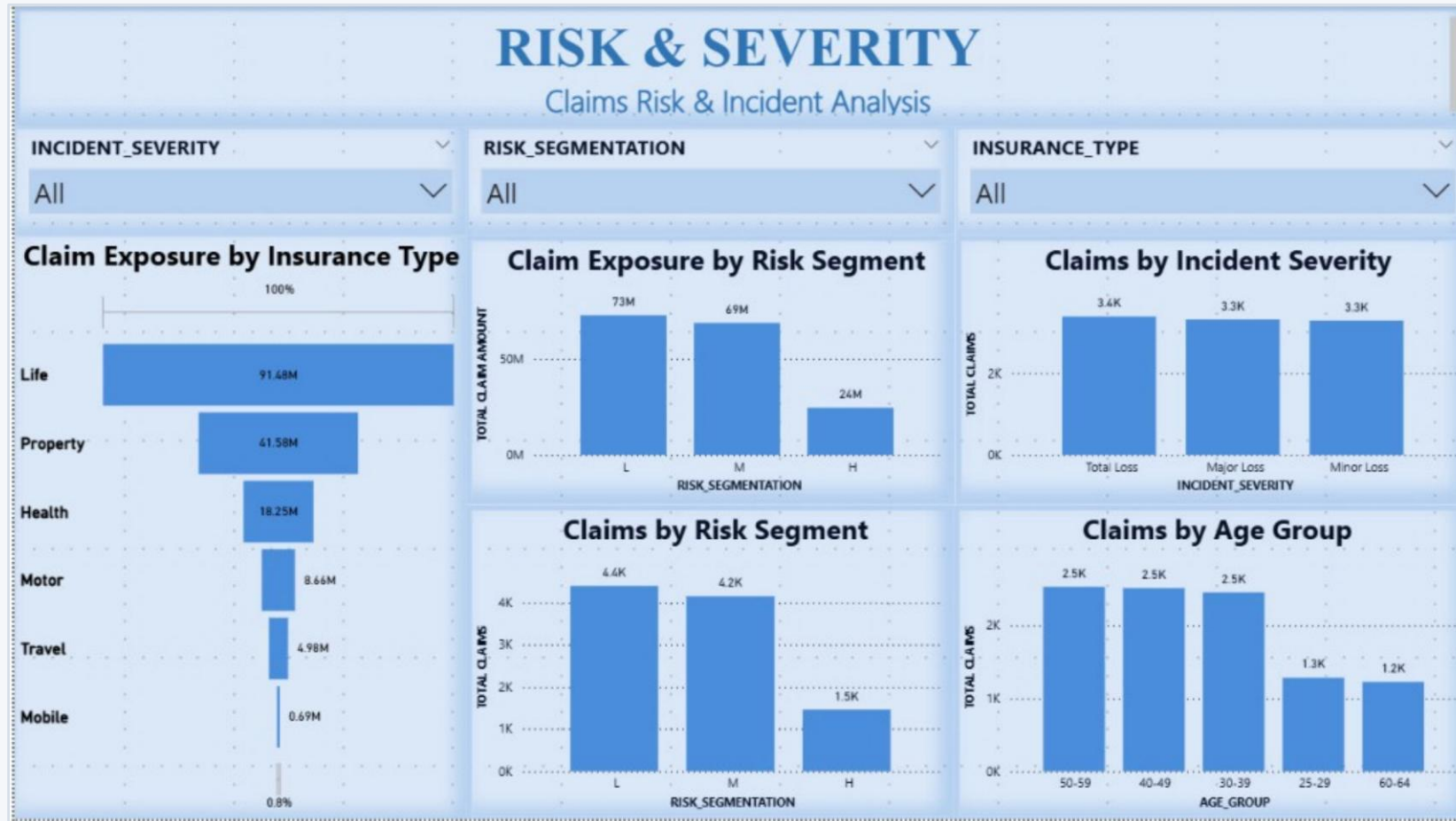
Provide a high-level view of claim volume, financial exposure, insurance type distribution and monthly claim trends.

Key Components

- Total Claims
- Total Claim Amount
- Average Claim-to-Premium
- High-Value Claims
- Same-Day Claims
- Monthly Claim Amount Trend
- Claim Exposure by Insurance Type
- Claim Volume by Insurance Type
- Risk & Insurance Type slicers

This page provides the overall business-level view of insurance claim activity and financial exposure.

Power BI Dashboard 02 — Risk & Severity



Purpose

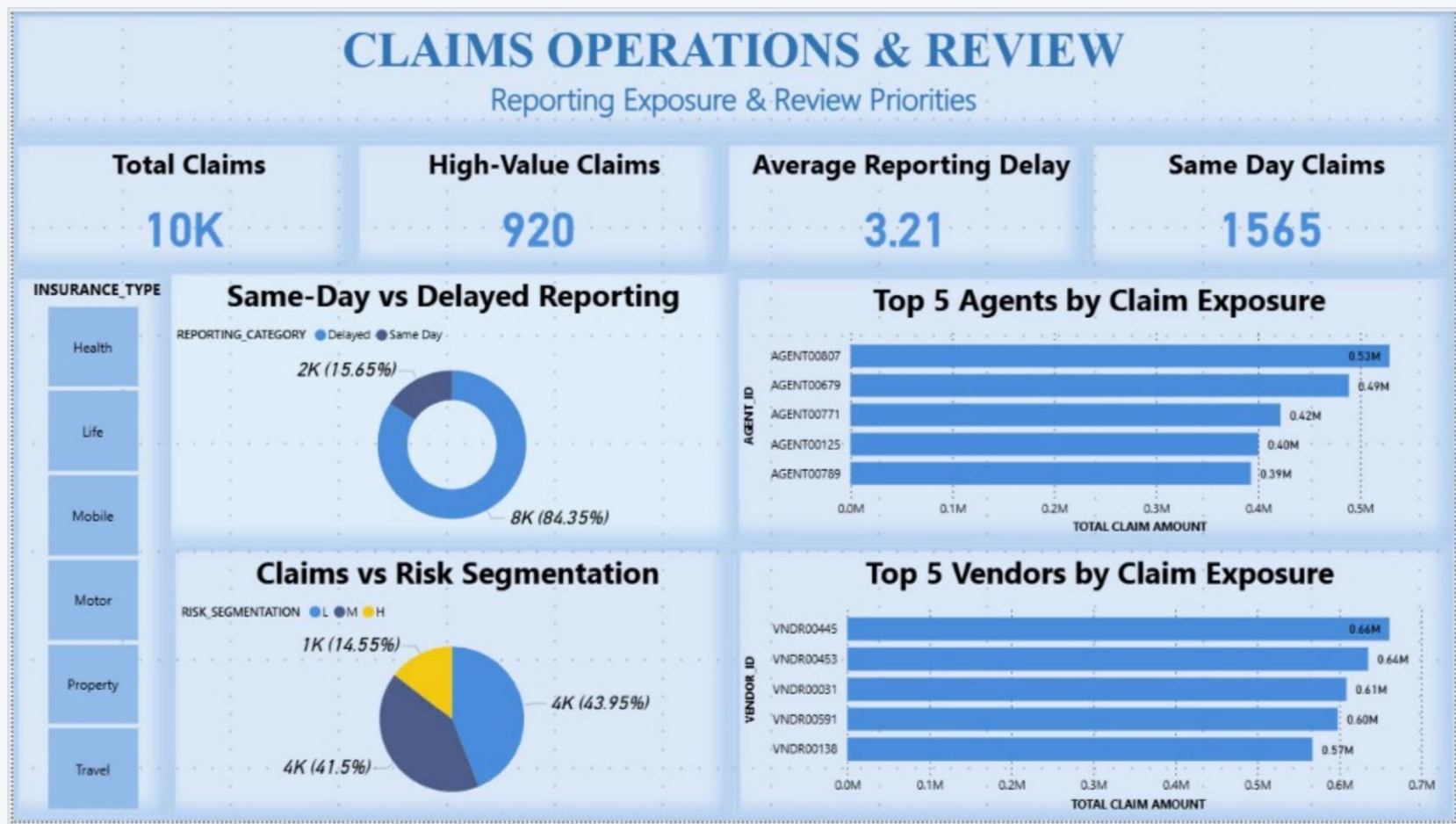
Understand the distribution of risk segments, incident severity, claim exposure and customer age groups.

Key Components

- Claims by Risk Segment
- Claims by Incident Severity
- Claim Exposure by Risk Segment
- Claims by Age Group
- Risk × Incident Severity matrix
- Insurance Type slicer
- Risk Segment slicer
- Incident Severity slicer

Low and Medium risk segments account for most claims, while High risk represents 14.55% of total claims.

Power BI Dashboard 03 — Claims Operations & Review



Purpose

Analyze reporting behavior and operational claim exposure across agents and vendors.

Key Components

- Total Claims
- High-Value Claims
- Average Reporting Delay
- Same-Day Claims
- Same-Day vs Delayed Reporting
- Top Agents by Claim Exposure
- Claims by Risk Segmentation
- Top Vendors by Claim Exposure

84.35% of claims are reported after the loss date, with an average reporting delay of 3.21 days.

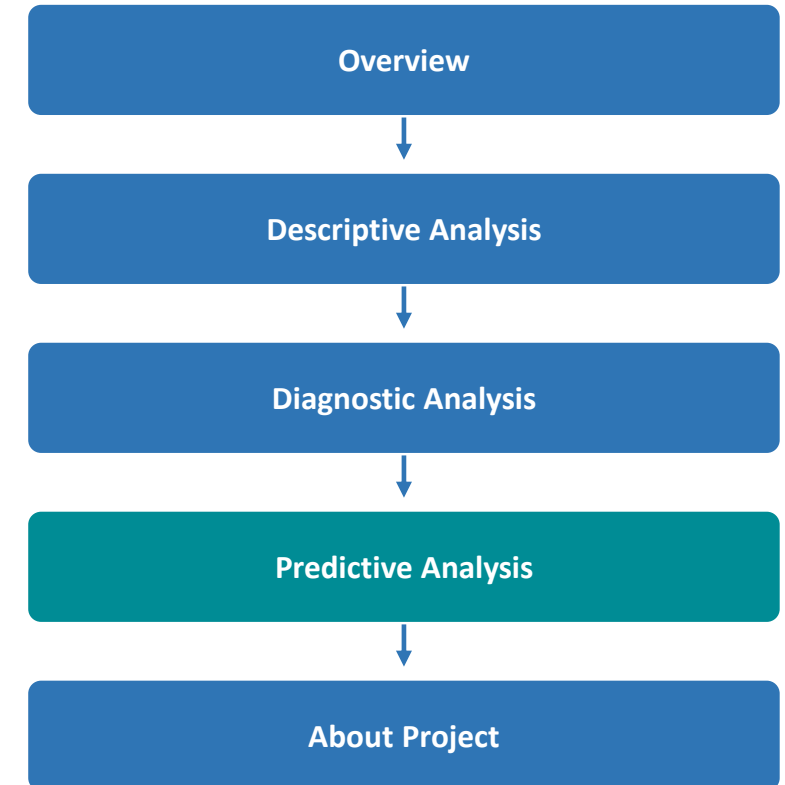
Streamlit Analytics Application

The Streamlit application provides an interactive Python-based analytics interface connected directly to the Snowflake analytical view.

Application Features

- Snowflake data connection
- Secure credentials via Streamlit secrets
- Cached data loading
- Interactive filters
- Dynamic KPIs
- Descriptive analysis
- Diagnostic analysis
- Predictive analysis
- ML prediction interface

Application Navigation



Screenshot of the running application is included in the project repository.

Key Business Insights

1 Financial Exposure

10,000 claims with total exposure of ₹165.64M. Average claim is ₹16,563.83 while the median is ₹7,000 — a right-skewed distribution.

2 Risk Distribution

Low-risk claims account for 4,395, Medium-risk for 4,150 and High-risk for 1,455 (14.55% of the book).

3 Incident Severity

Total Loss represents 3,390 claims, Major Loss 3,317 and Minor Loss 3,293 — an almost even split.

4 Reporting Behaviour

Average reporting delay is 3.21 days. 1,565 claims were reported on the same day as the loss date.

5 Data Quality

No full-row duplicates and no invalid agent or vendor relationships were identified during validation.

6 Predictive Modeling

Linear Regression achieved an R^2 of 0.7066 on the test set, explaining a substantial share of claim-amount variation.

Business Recommendations

01

Monitor Claim Exposure

Track high-value claims and claim exposure by insurance type to support financial planning and operational monitoring.

02

Risk-Based Monitoring

Use risk segmentation together with incident severity to identify claim groups requiring closer operational review.

03

Reporting Monitoring

Monitor reporting delays and same-day reporting patterns to understand operational reporting behaviour.

04

Agent & Vendor Monitoring

Use claim volume and financial exposure metrics to monitor operational workload across agents and vendors.

05

Predictive Analytics

Use claim amount prediction as a supporting analytical tool for estimation and planning, with continuous validation against future outcomes.

Project Deliverables

All assignment components completed end to end



Excel Data Profiling

Completed



Snowflake Setup

Completed



SQL Transformation

Completed



Analytical Views

Completed



Databricks Analysis

Completed



Power BI Dashboards

Completed



Project Presentation

Completed



Excel Data Quality Analysis

Completed



SQL Data Cleaning

Completed



SQL EDA

Completed



Python Analysis

Completed



Machine Learning

Completed



Streamlit Application

Completed



GitHub Repository

Completed

Conclusion

This project demonstrates an end-to-end insurance claims analytics workflow, from raw data profiling and quality assessment through SQL-based data preparation, Python analysis, business intelligence and machine learning.

Raw claims data was transformed into a structured analytical dataset and delivered through Power BI and Streamlit.

The analysis provides visibility into claim exposure, insurance types, risk segments, incident severity, customer characteristics, reporting behaviour, agent and vendor activity, and claim amount prediction.

The final solution demonstrates how multiple data analytics tools work together to support a complete business analytics workflow.



THANK YOU

Appendix A — Insurance Data Dictionary

Key insurance fields with business meaning and data type

Field	Business Meaning	Type
TRANSACTION_ID	Unique claim transaction identifier	Identifier
CUSTOMER_ID	Customer identifier	Identifier
POLICY_NUMBER	Policy identifier	Identifier
TXN_DATE_TIME	Transaction timestamp	Date/Time
POLICY_EFF_DT	Policy effective date	Date
LOSS_DT	Date of loss	Date
REPORT_DT	Date of reporting	Date
INSURANCE_TYPE	Insurance category	Categorical
CLAIM_AMOUNT	Claim amount	Numeric
PREMIUM	Premium amount	Numeric
AGE	Customer age	Numeric
TENURE	Customer / policy tenure	Numeric

Field	Business Meaning	Type
RISK_SEGMENTATION	Risk category	Categorical
INCIDENT_SEVERITY	Incident severity category	Categorical
ANY_INJURY	Injury indicator	Categorical
POLICE_REPORT_AVAILABLE	Police report indicator	Categorical
AGENT_ID	Agent identifier	Identifier
VENDOR_ID	Vendor identifier	Identifier
MARITAL_STATUS	Marital status	Categorical
EMPLOYMENT_STATUS	Employment status	Categorical
HOUSE_TYPE	House type	Categorical
SOCIAL_CLASS	Social class	Categorical
CUSTOMER_EDUCATION_LEVEL	Education level	Categorical

Appendix B — Derived Analytical Fields

Fields engineered during SQL transformation

1

REPORTING_DELAY_DAYS

Days between loss date and report date

2

CLAIM_TO_PREMIUM_RATIO

Claim amount relative to premium

3

AGE_GROUP

Customer age grouping

4

TENURE_GROUP

Tenure grouping

5

CLAIM_AMOUNT_BUCKET

Claim amount range category

6

REPORTING_CATEGORY

Reporting delay category

7

CLAIM_VALUE_CATEGORY

Claim value category

8

LOSS_YEAR

Year extracted from loss date

9

LOSS_MONTH

Month extracted from loss date

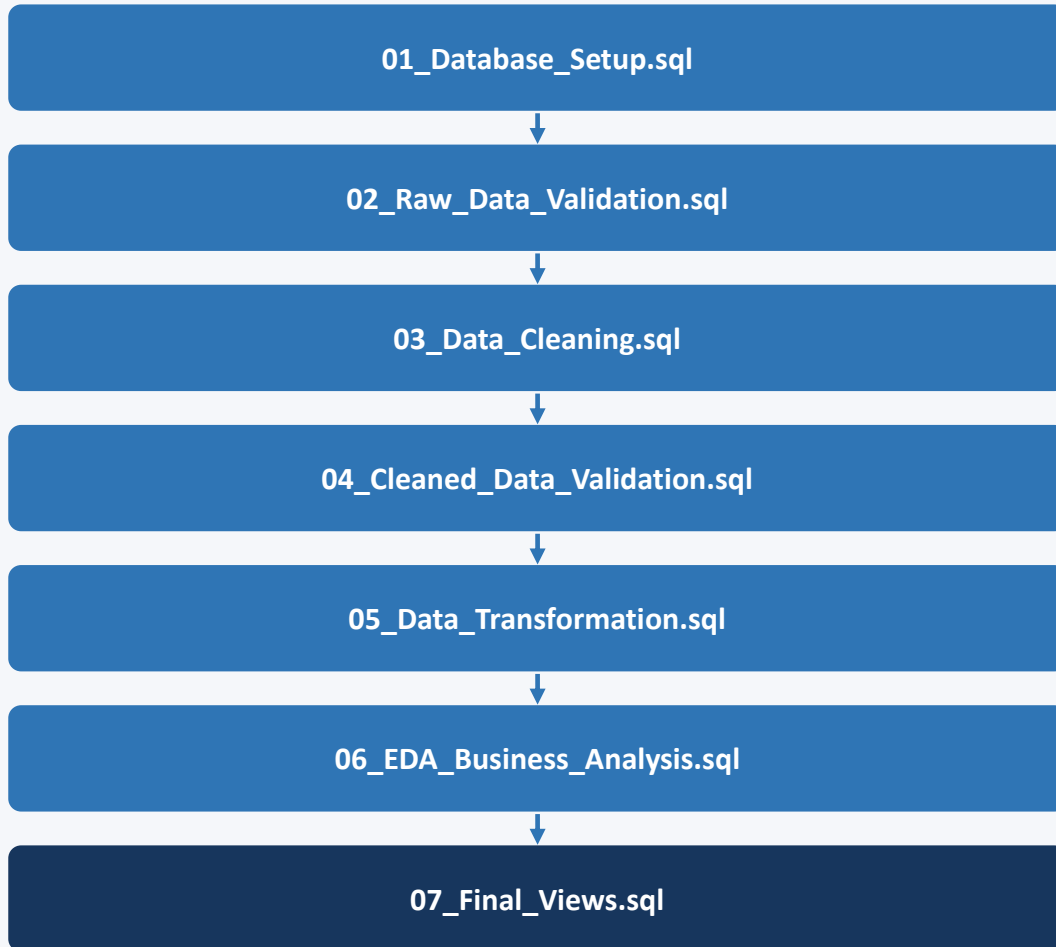
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LOSS_MONTH_NAME

Month name extracted from loss date

Appendix C — SQL Processing Pipeline

Execution Order



Final Analytical Views

- CLAIMS_ANALYTICS_VW
- CLAIMS_SUMMARY_VW
- INSURANCE_TYPE_VW
- RISK_SEGMENT_VW
- INCIDENT_SEVERITY_VW
- AGENT_PERFORMANCE_VW
- VENDOR_PERFORMANCE_VW
- MONTHLY_CLAIMS_TREND_VW
- STATE_CLAIMS_VW
- CLAIM_INVESTIGATION_PRIORITY_VW

Appendix D — Key Power BI Measures

DAX measures built on CLAIMS_ANALYTICS_VW

Volume Measures

- Total Claims
- Total Customers
- Total Agents
- Total Vendors
- High-Value Claims

Financial Measures

- Total Claim Amount
- Average Claim Amount
- Median Claim Amount
- Total Premium
- Average Premium

Ratio & Operational

- Average Claim-to-Premium Ratio
- Average Reporting Delay
- Same-Day Claims
- Same-Day Reporting %
- High-Value Claim %